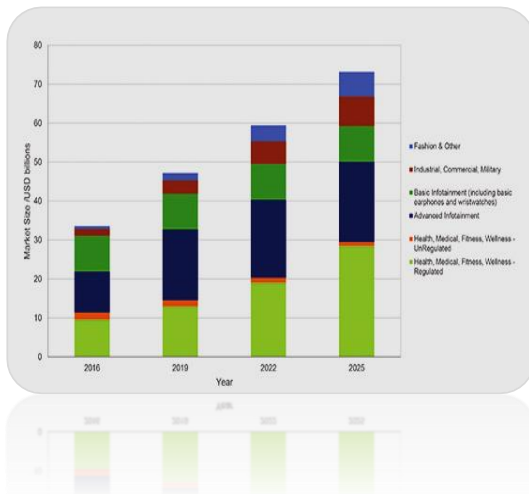


Graphene is on the move

Dr Khasha Ghaffarzadeh

IDTechEx

Research



- Technology landscape
- Company profiles
- Market forecasts

Consulting



- Custom research
- Analyst access
- Advisory service

Show!



- Conferences
- Exhibition
- Masterclasses

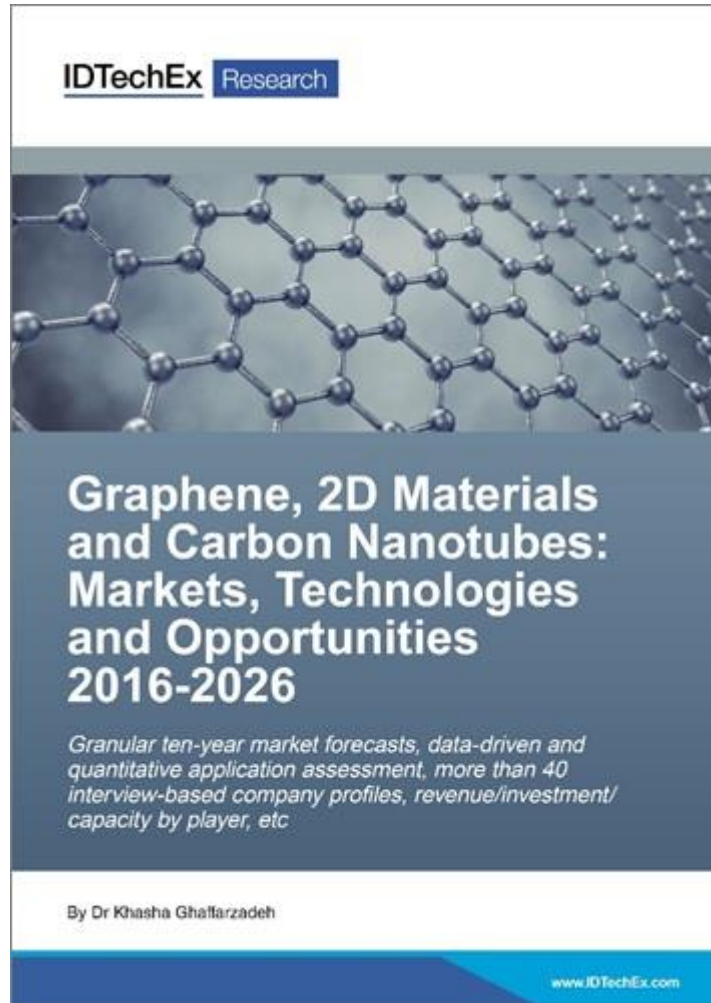
Why IDTechEx?

We are **technical, impartial** and **experienced** in our chosen research topics.

- **Globally cited** analyst team combining technical analysts (most have PhDs) and successful business leaders.
- **Global Research**: in 2014 we visited China, Japan, USA, Korea, Germany, Brazil, Belgium, France, UK, The Netherlands, Canada, etc
- **Hundreds of interviews** are conducted each year as part of our global research programmes with organisations across the value chain
- **80,000+ global contacts** in our targeted contact database of the printed, organic and flexible electronics industry. More than **20,000** people read our journal www.PrintedElectronicsWorld.com

Graphene, CNTs and 2D Materials

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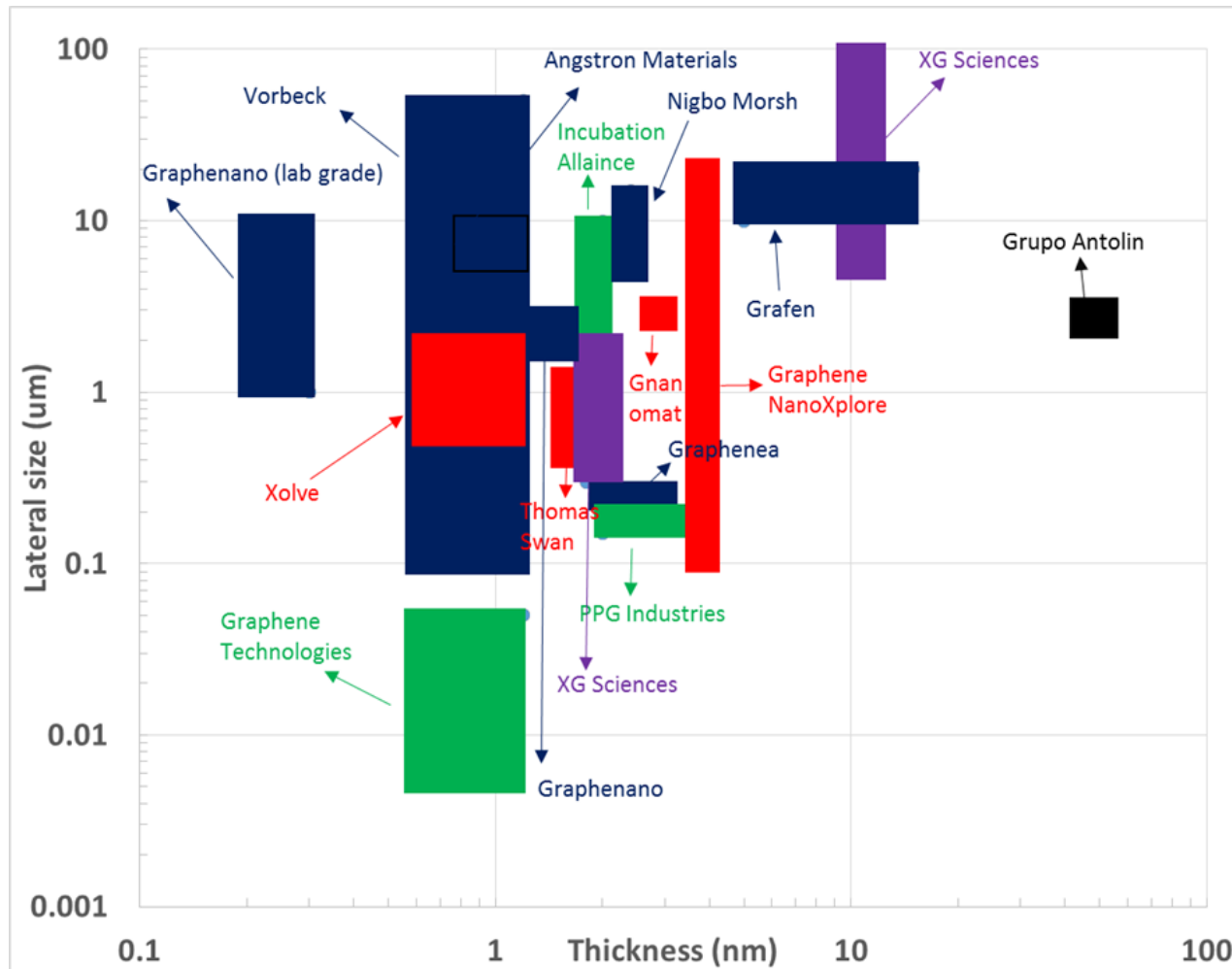


Graphene, CNTs and 2D Materials 2015:2025

www.IDTechEx.com/graphene

- Detailed technology description
- >35 interview-based company profiles
- Production capacity globally
- Value proposition, progress and challenges analysed for each target market
- Ten-year market forecasts segmented by application and graphene type
- More 18 sectors analysed in detailed

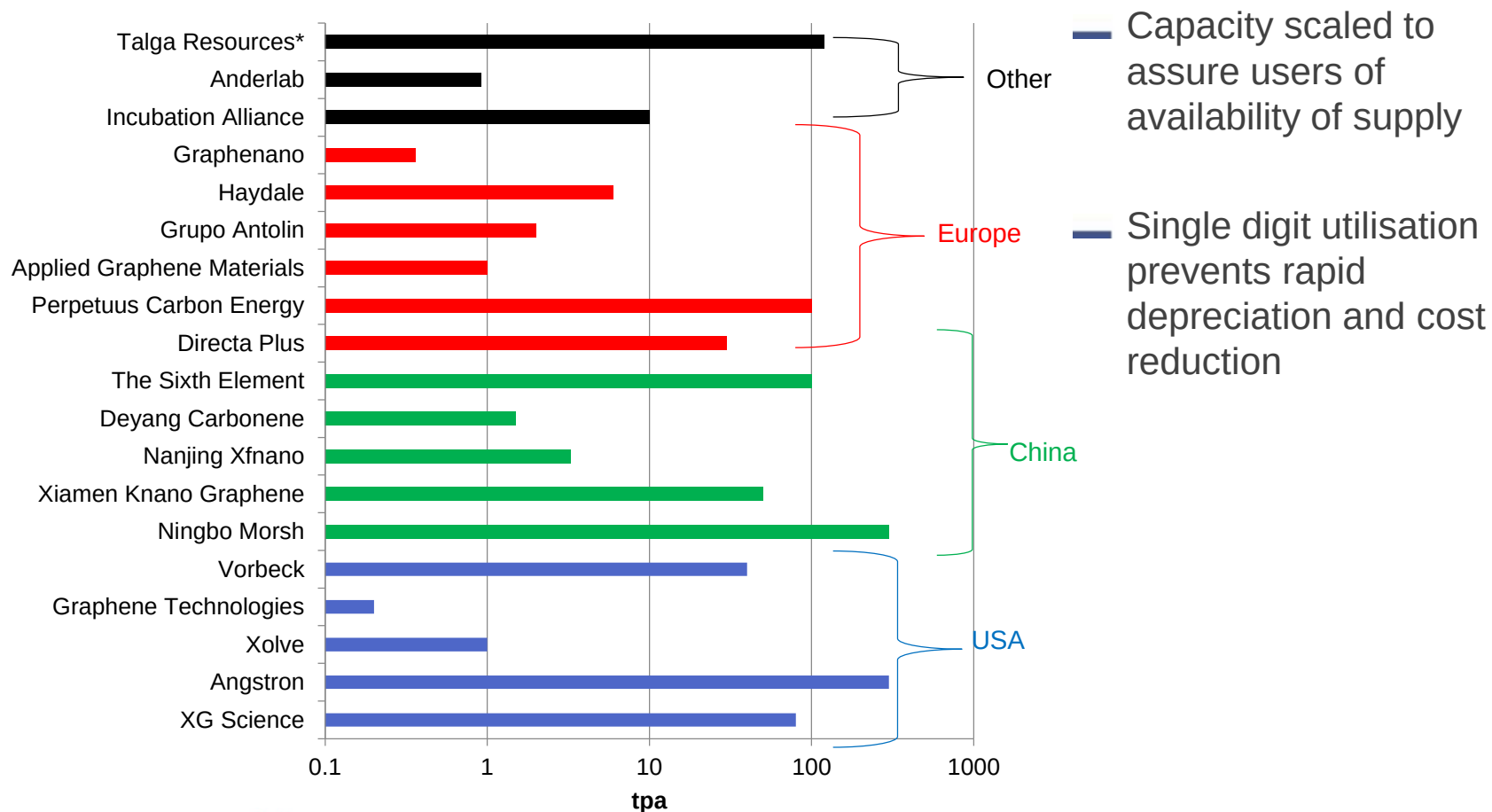
Not all graphene platelets are the same



Source: IDTechEx

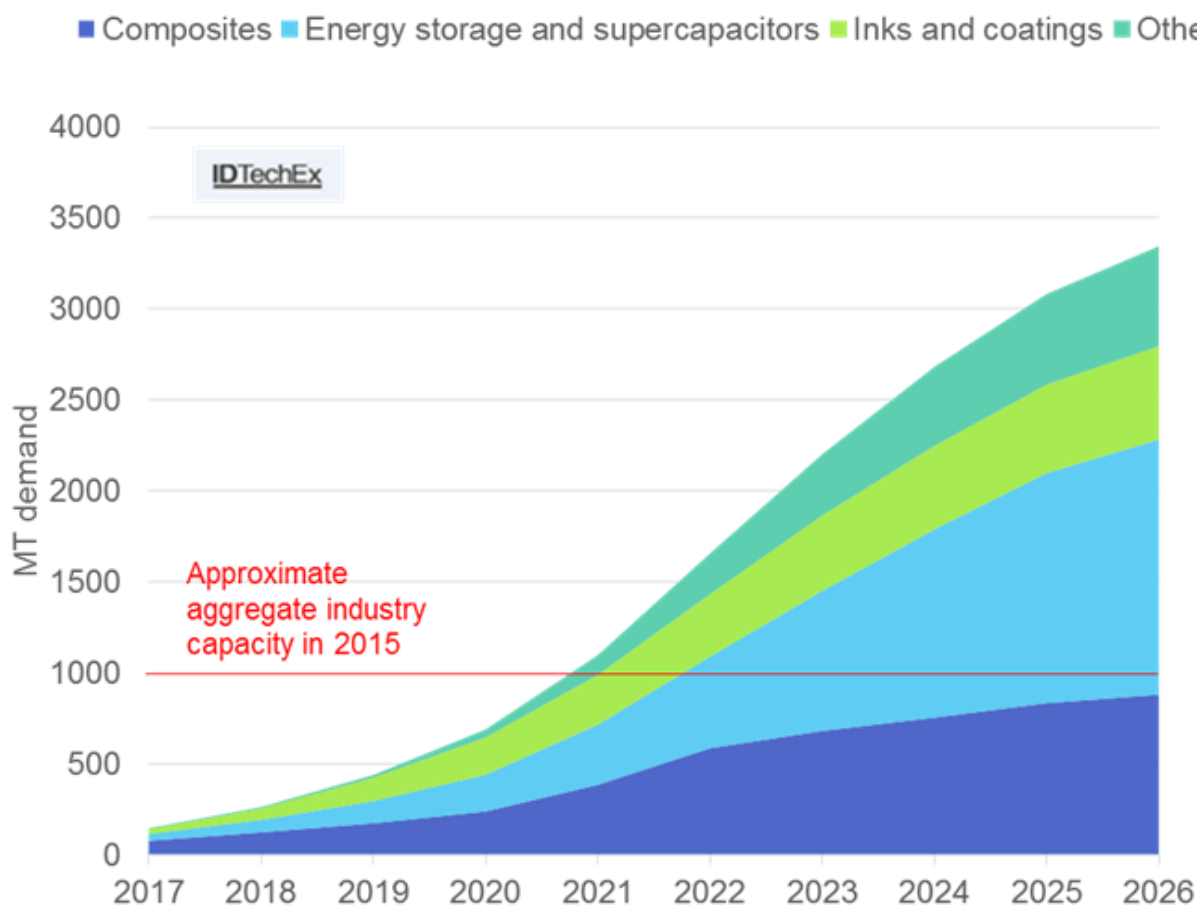
Making graphene available and affordable

IDTechEx



*Talga's capacity is not online yet- It is based on their announced plans

Making graphene available and affordable



- Graphene prices falling faster than previously anticipated
- Graphene still over-priced
- Availability is no longer a major impediment for adoption
- We have reduced our total revenue forecast, but substantial increased our volume adoption



Intermediaries will unblock the market

Graphene

Company	Ink /Slurry	Masterbatch/ Composites
XG Science	Yes	Yes
Vorbeck	Yes	Yes
Haydale	Yes	Yes
Xolve		Yes
Angstron	Yes	Yes
Ningbo Morsh	Yes	Yes
KNano	Yes	Yes
Anderlab	Yes	Yes
Graphene Technologies	Yes	

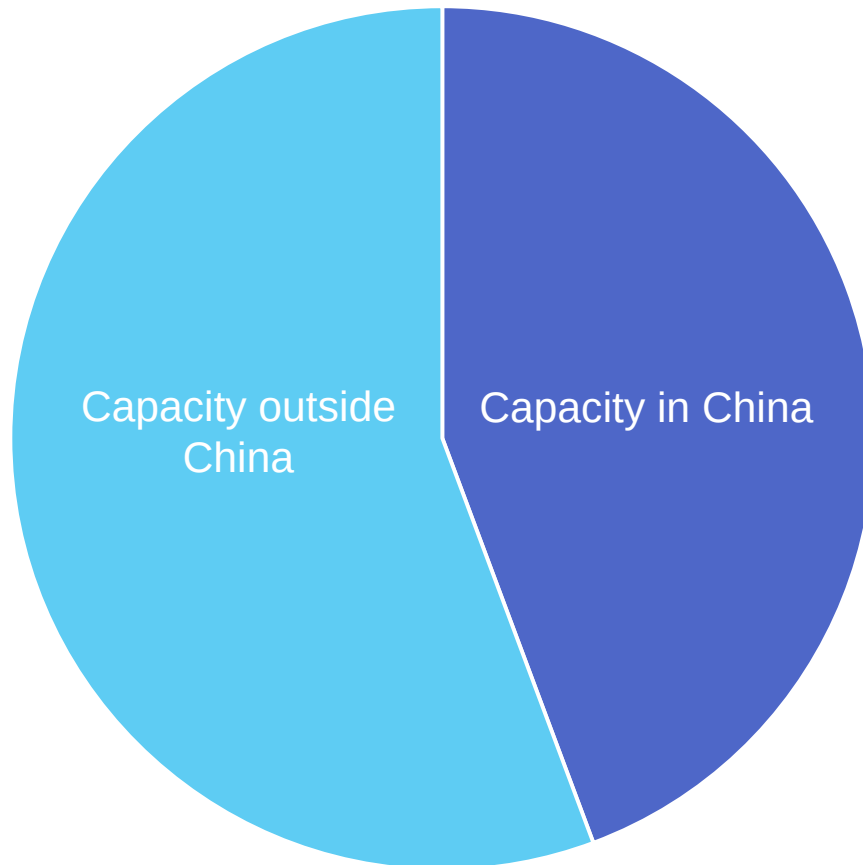
CNTs

Company	Masterbatch/ Composites
Arkema	Yes
Nanocyl	Yes
Hyperion	Yes
Ocsial/Zyvex	Yes
Hanhwa Chemical	Yes
Shenzen Nanotech	Yes
MK Nano	Yes
Timesnano	Yes

- The trend in the industry is to offer intermediaries in order to facilitate or ‘unblock’ the market.
- The table on the right shows graphene companies that have moved downstream (closer to customers) to offer intermediaries
- We note that a similar trend took place with the CNT industry. Many companies moved downstream and some decided only to offer intermediaries

Centre of graphene gravity shifts to China?

Share of production capacity in 2015

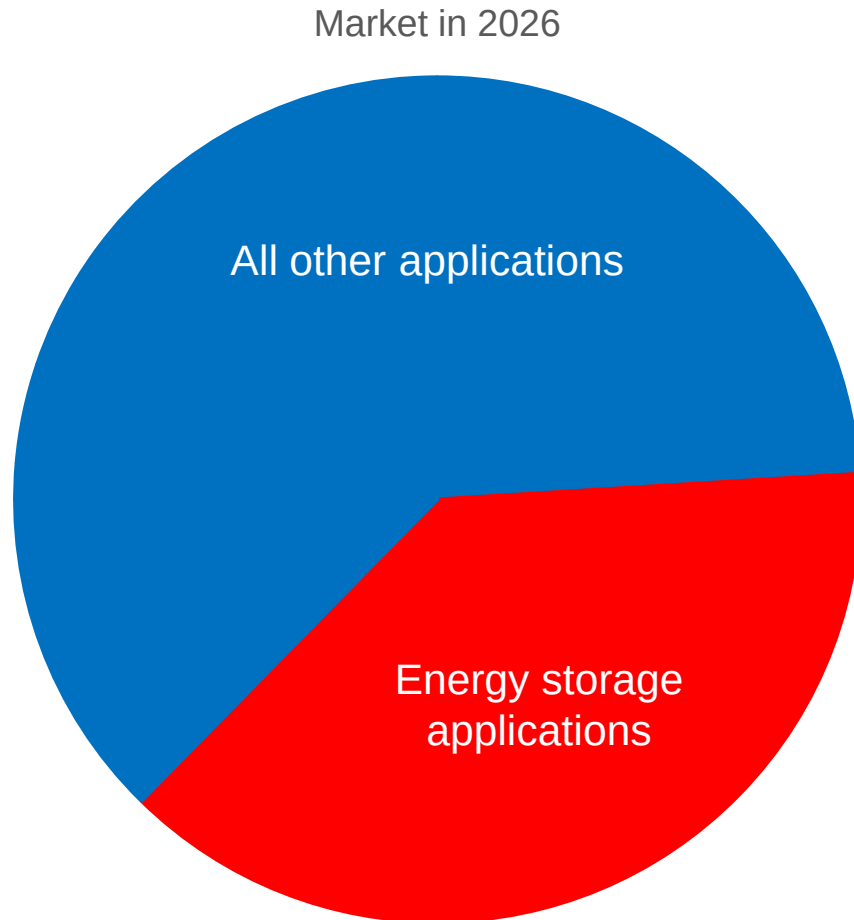


- Chinese entities are leaving behind no whitespace in the patent landscape
- Ambitious production and pricing plans for both CVD graphene and platelet-type graphene companies
- Chinese firms also found success in the CNT world



■ 1 ■ 2

Pivot towards energy storage applications



- The application divergence period is coming to an end, and energy storage is emerging as a focus area
- Energy storage electrodes are carbon based
- Graphene improves the electrode resistance, preventing performance degradation at high C

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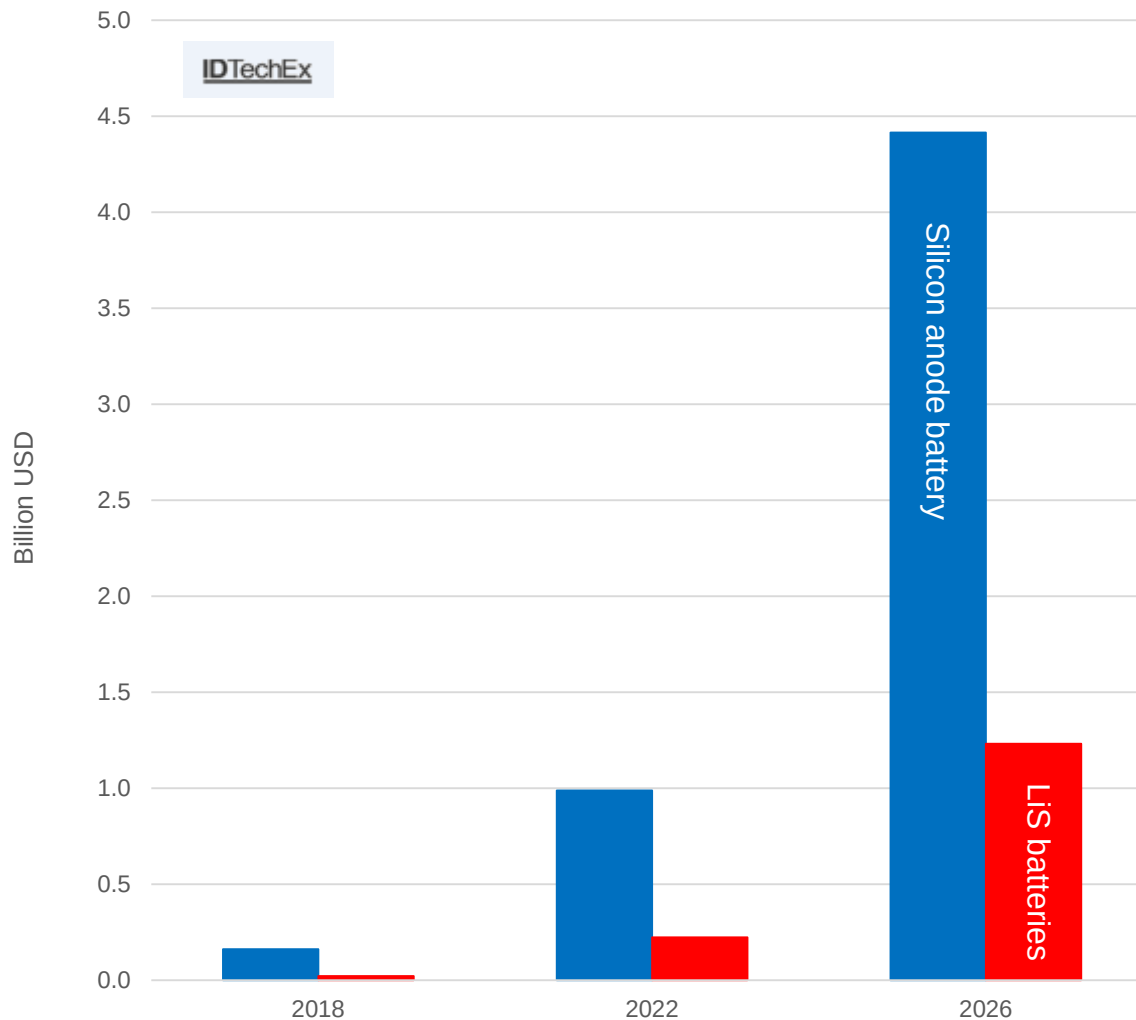
CABOT

SION
POWER

OXIS ENERGY
Next Generation Battery Technology

GRAFOID

Pivot towards energy storage applications



— New battery technologies bring big potential but also big problems: opportunity for innovation

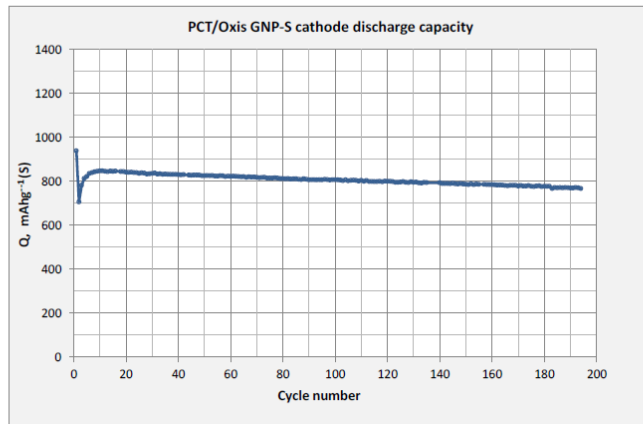
— Graphene and graphene oxide can help improve the lifetime, one of the critical challenges facing the industry



Silicon/graphene battery electrodes

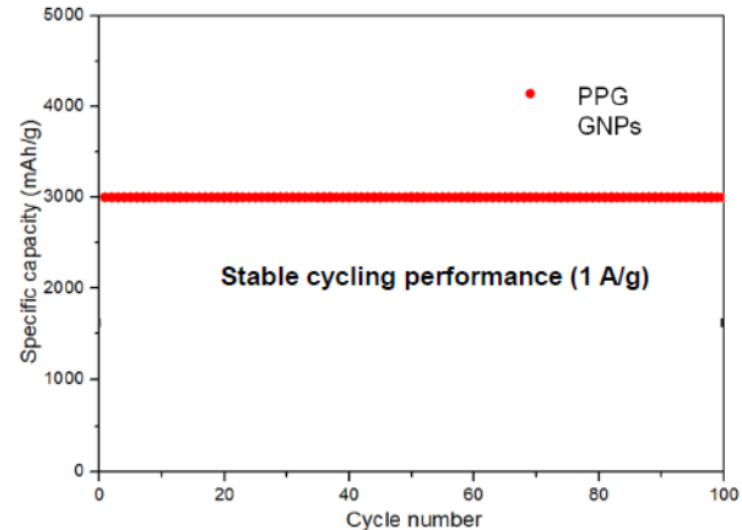
Graph showing Discharge capacity vs cycle number for a PCT/Oxis cell using Stacked Graphenes S intercalated composite cathodes.

$$I_D = C/5; I_C = C/10$$



Source: Perpetuus and Oxis Energy

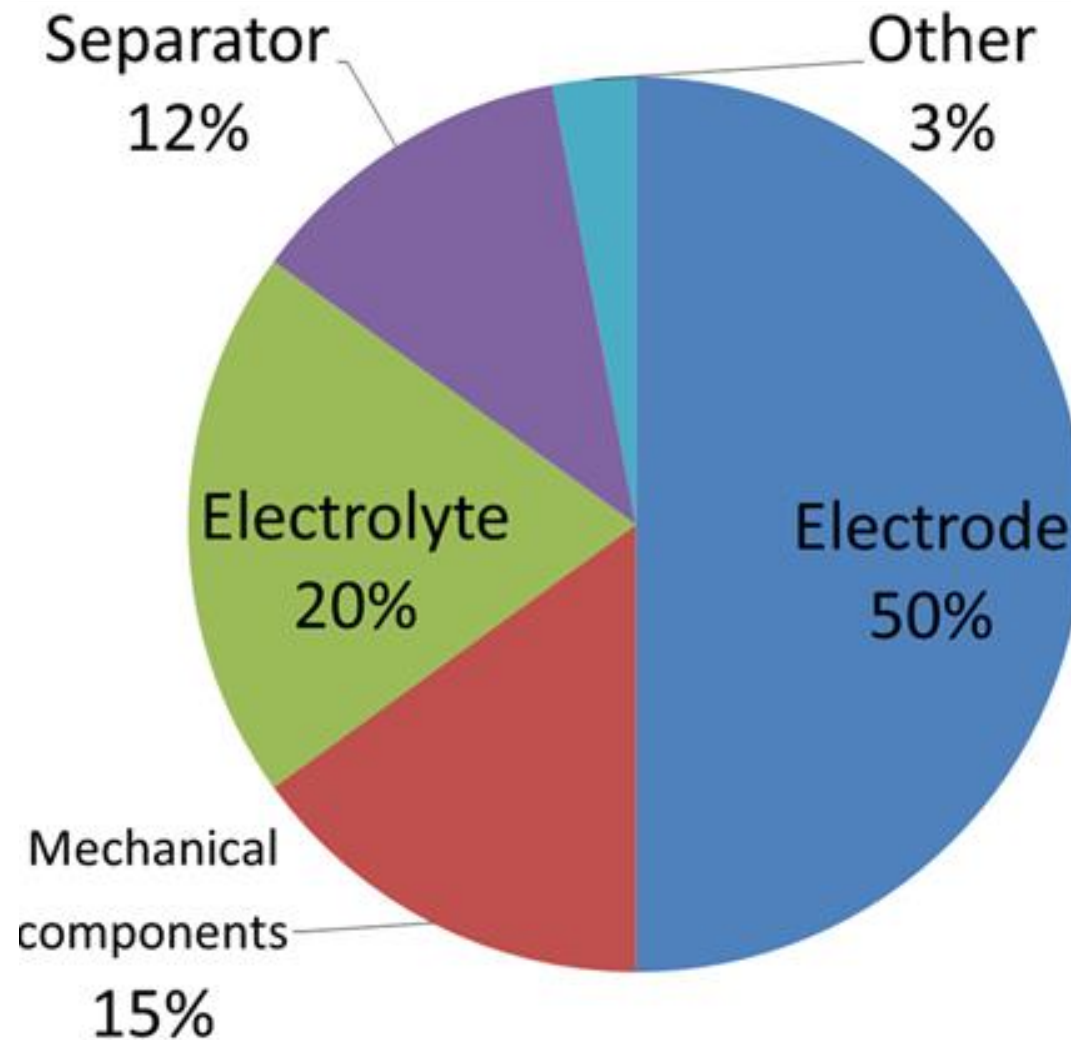
Capacity-limited cycling
Si/PPG GNPs



Source: PPG Industries

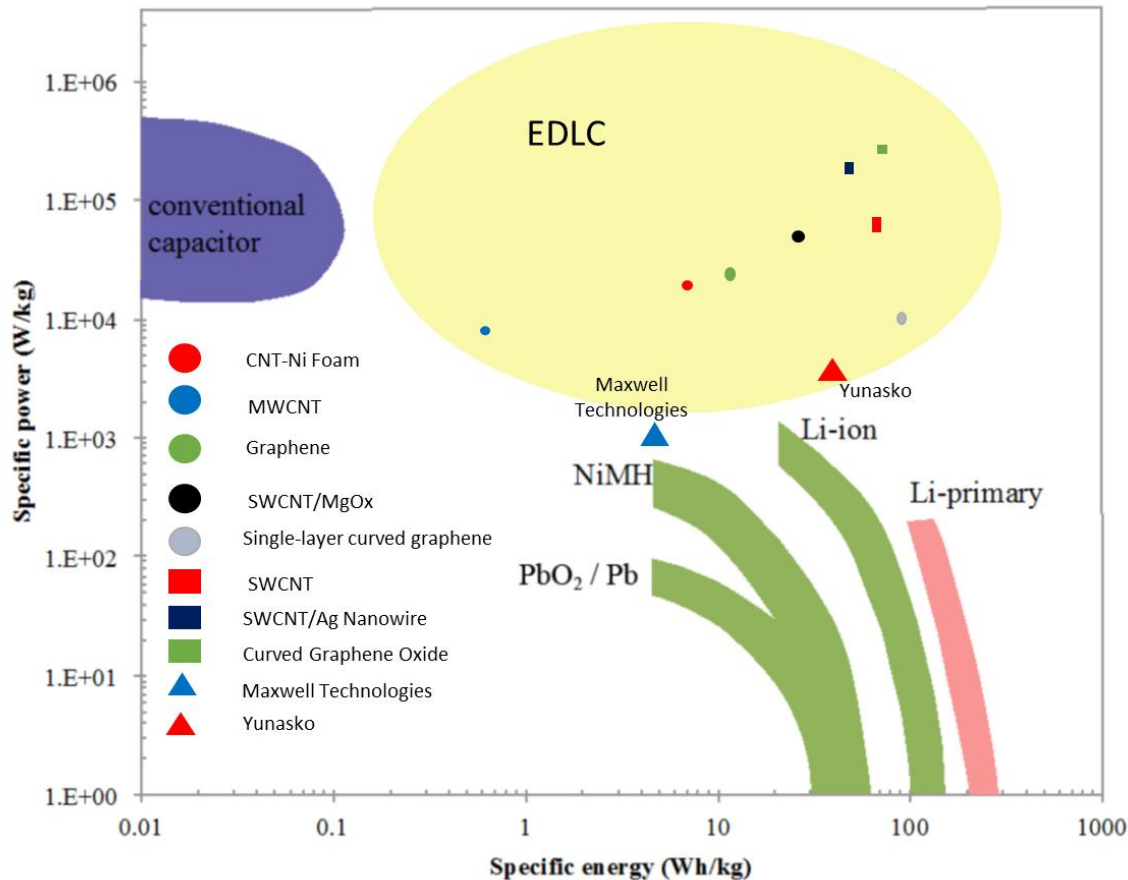
- A major challenge for silicon anode batteries is managing volumetric changes of silicon. Graphene can act as a buffer layer to absorb some of the strain thus helping extend the lifetime
- A major challenge for lithium sulphur is in retaining the active material and preventing from leaking into the electrolyte during the charge up and discharge up process

Electrodes represent a major cost of supercaps



Source: IDTechEx

Graphene improves the performance of supercapacitors?



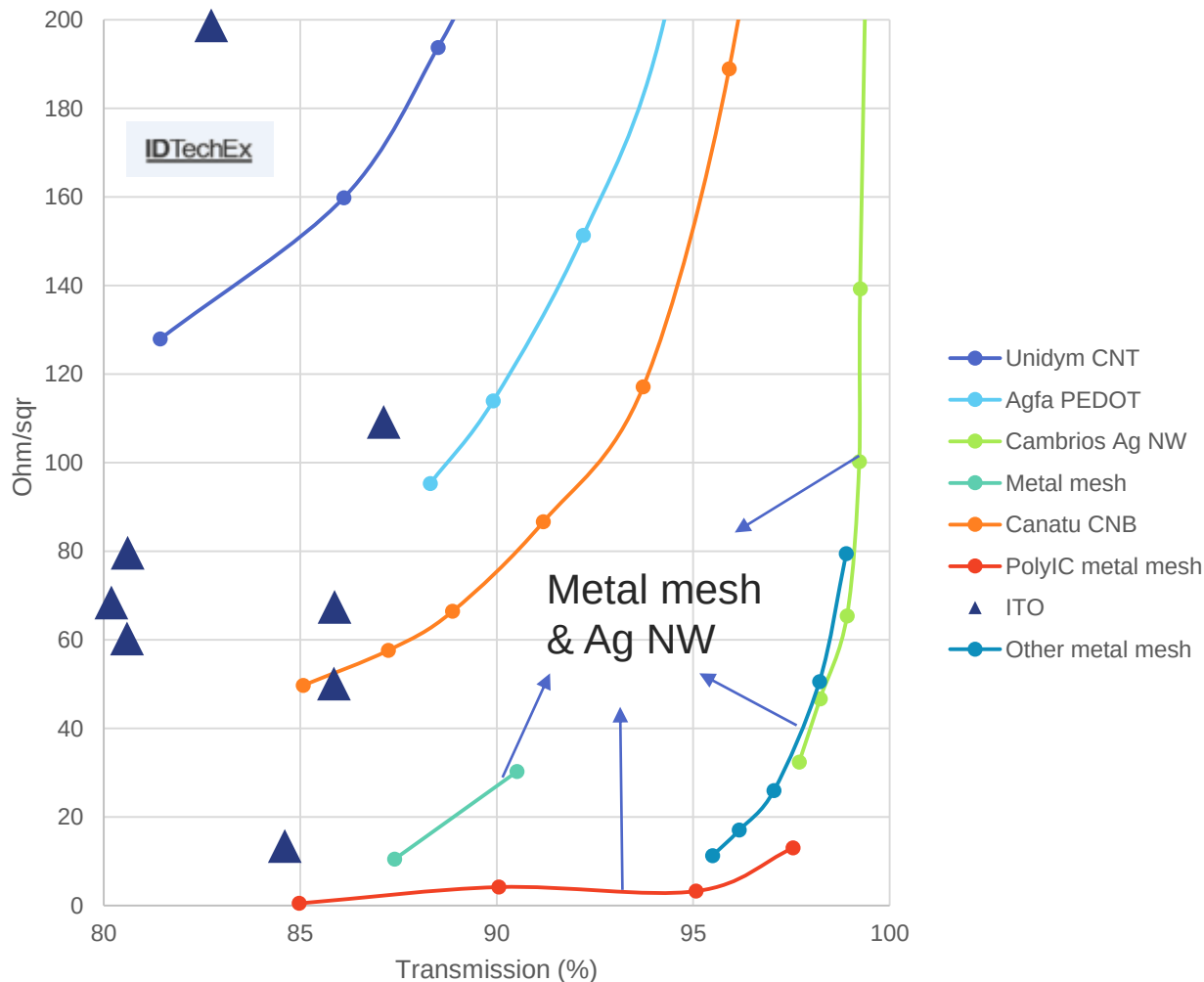
— Nano carbon and nanocarbon hybrids are pushing the performance boundaries of supercapacitors, as shown in this chart which compares the latest research with commercial products

— Curved graphene oxide is amongst the best performers.

— Graphene is curved to prevent re-staking

Source: IDTechEx and various

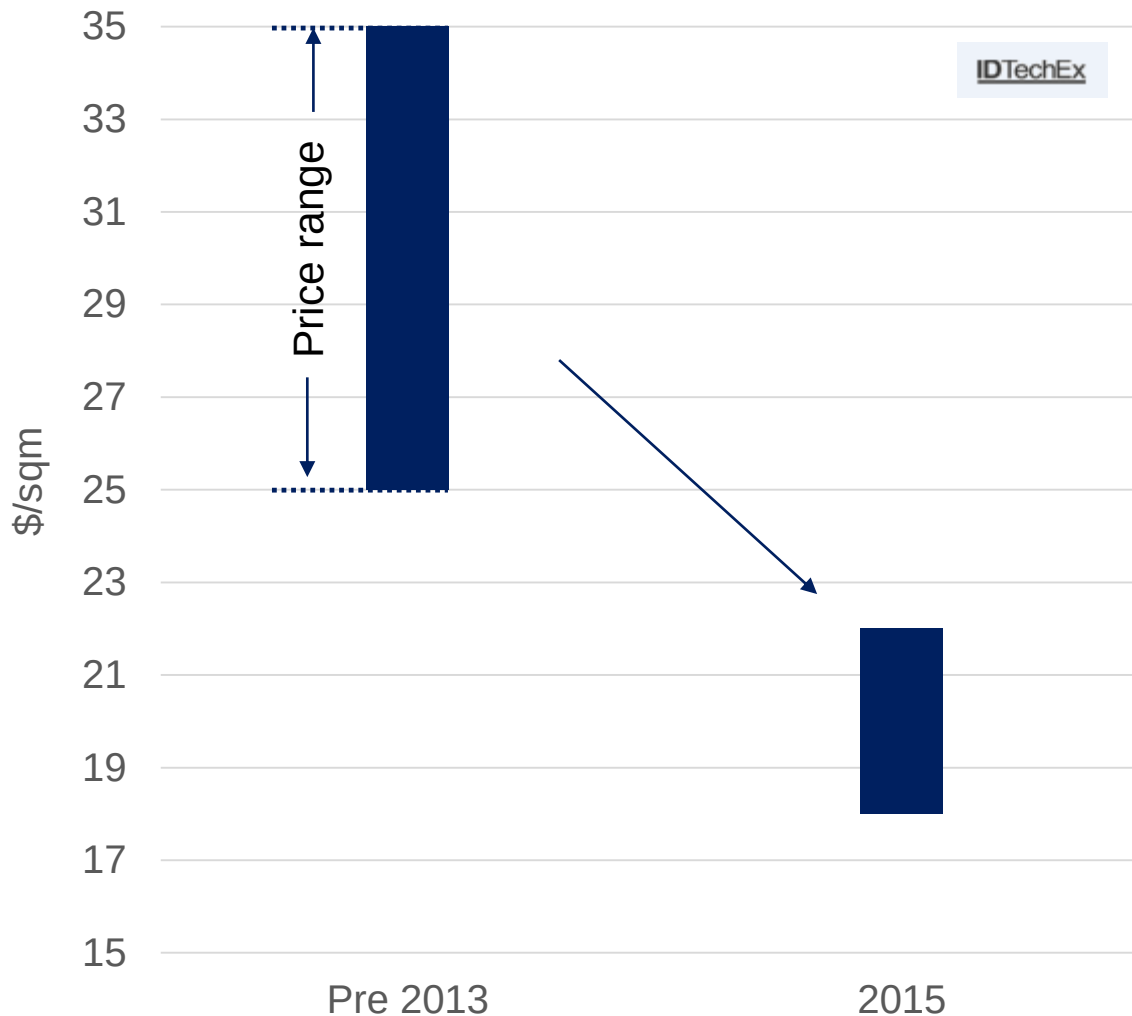
Graphene TCFs are not good enough



Graphene offers mediocre performance at best

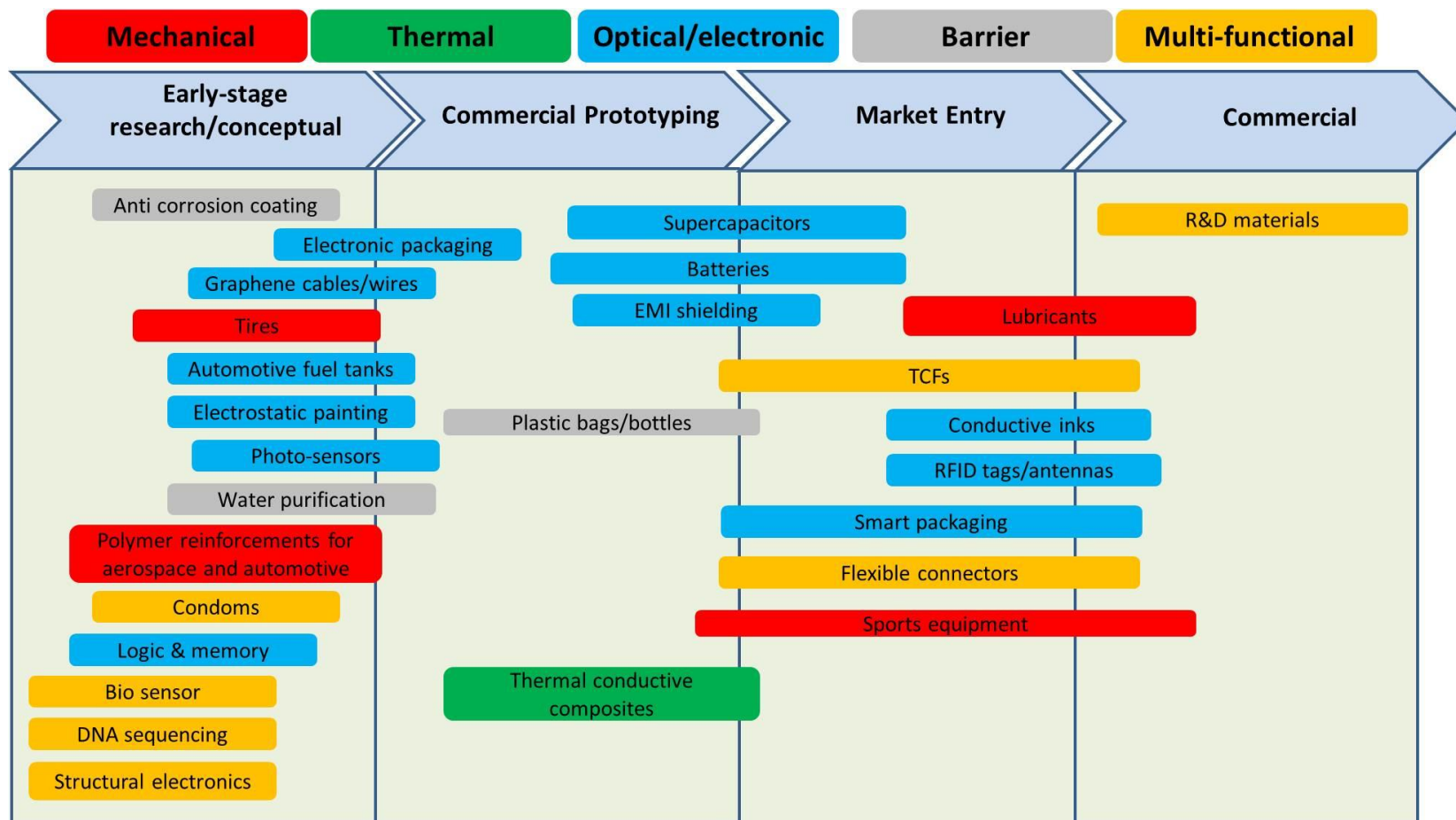
Questions remain over cost, yield and scalability of the transfer process

Price erosion triggers consolidation

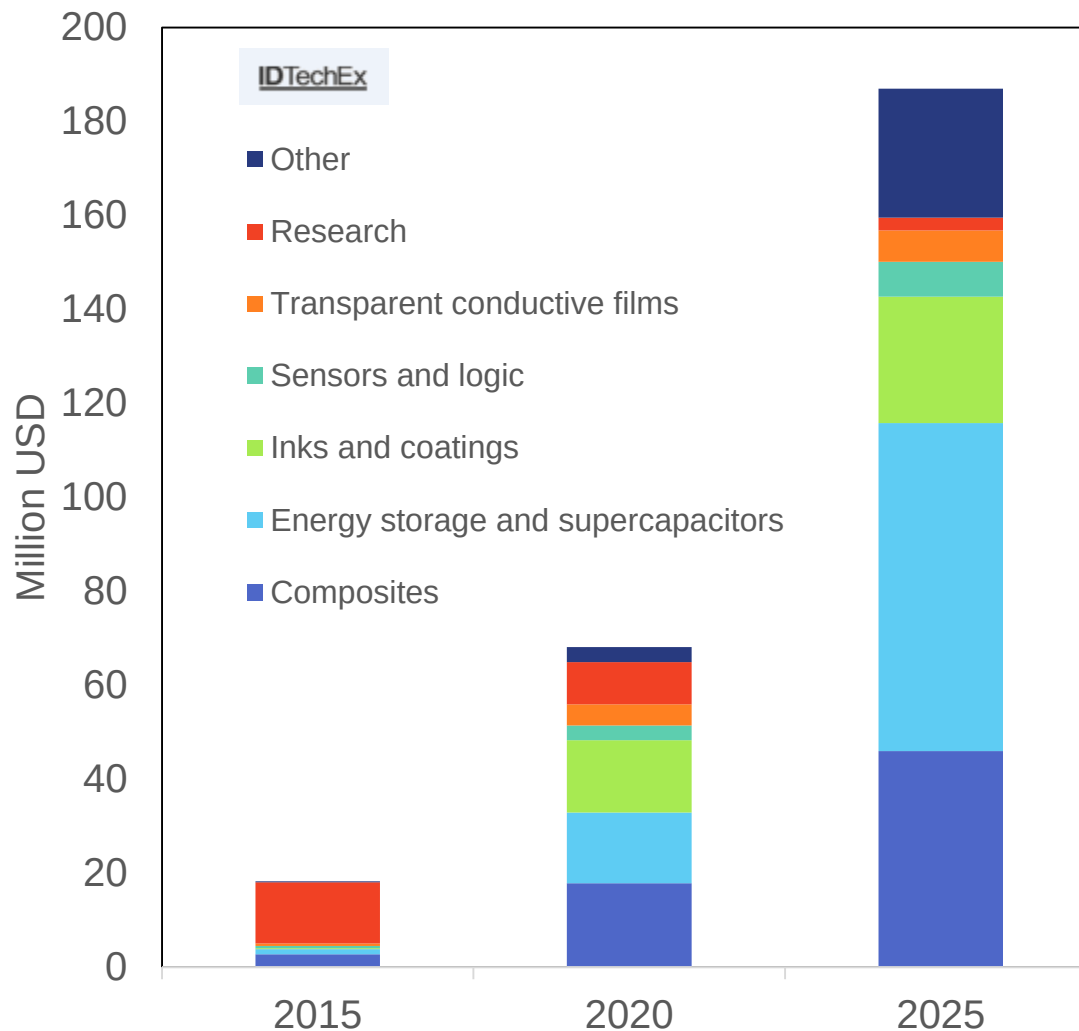


- Price and margin erosion driven by the threat of substitutes
- Incumbents reduce prices piece-meal every time they are about to lose a business/project.
- The strategy is to protect market share and make life hard on alternatives

Application timeline



Insights and Forecasts



- No 'killer application' yet
- 'Substitution' strategy against well-entrenched incumbents
- Following trajectory of CNTs?
- Many target markets thus high probability of success but also opportunity overdose for small firms
- Small gain on multiple fronts
- More intermediary products-dispersion challenge?
- Need an 'other' category?